Mahdi Kallel

PhD candidate in reinforcement learning

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Profile

- PhD candidate in reinforcement learning at the University of Würzburg (graduating April 2027), with a background in applied mathematics and optimization. I began my PhD on policy optimization and exploration, which led me to the networks underneath. My research now focuses on the learning dynamics of neural networks, in particular how reinforcement learning and supervised learning differ in plasticity and generalization.

Publications

The Spectral Anatomy of Plasticity Loss 2026

Used random matrix theory to split each weight spectrum into a random bulk and learned edges, and tracked it over a long stream of continual tasks. Directions learned on earlier tasks capture a growing share of new gradients. Making them read-only keeps the network trainable while preserving what it already learned.

Mahdi Kallel, Nour Jamoussi, Johannes Tölle, Marios Kountouris, Carlo D'Eramo

www.eurecom.fr/en/publication/8988 (AXIOM Workshop at NeurIPS 2026)

Do Not Imitate, Reinforce: Iterative Classification via Belief Refinement 2026

Replaced label imitation with reinforcement learning, treating classification as sequential belief refinement where a recurrent agent is rewarded for improving its prediction. The classifiers match supervised accuracy and stay calibrated under human label noise, where cross-entropy becomes overconfident. A value function decides when to stop.

Mahdi Kallel, Johannes Tölle, Ahmed Hendawy, Carlo D'Eramo

arxiv.org/abs/2604.22110 (Transactions on Machine Learning Research (TMLR), 2026)

PPO+: Enhancing Proximal Policy Optimization 2026

Traced the sensitivity of PPO to inaccurate critic estimates, and trained the critic on off-policy data while keeping the actor update on-policy. Better critics substantially improve PPO in continuous control and narrow the gap to off-policy algorithms, without relying on implementation tricks.

Mahdi Kallel, Jose-Luis Holgado-Alvarez, Samuele Tosatto, Carlo D'Eramo

rlj.cs.umass.edu/2026/papers/Paper8.html (Reinforcement Learning Journal (RLC 2026))

Use the Online Network If You Can: Towards Fast and Stable Reinforcement Learning 2026

Ahmed Hendawy, Henrik Metternich, Théo Vincent, **Mahdi Kallel**, Jan Peters, Carlo D'Eramo

arxiv.org/abs/2510.02590 (ICLR 2026)

Augmented Bayesian Policy Search 2024

Used the performance difference lemma to inject the critic into the Gaussian process prior of Bayesian optimization, bridging Bayesian optimization and policy gradient methods. Policy search with deterministic policies scales to high-dimensional locomotion tasks and outperforms earlier Bayesian optimization methods.

Mahdi Kallel, Debabrota Basu, Riad Akrour, Carlo D'Eramo

proceedings.iclr.cc/paper_files/paper/2024/hash/7eca17ef54789b0663cab421f2e9dbf5-Abstract-Conference.html (ICLR 2024)

Cyclicity-Regularized Coordination Graphs 2024

Oliver Järnefelt, **Mahdi Kallel**, Carlo D'Eramo

rlj.cs.umass.edu/2024/papers/Paper40.html (RLC 2024)

Research Experience

INRIA Lille, SCOOL Research Group, Research Intern — France, 2022
- Worked on Bayesian optimization for exploration with deterministic policies in continuous action spaces.

EURECOM, Research Intern — Biot, France, 2020
- Studied loss balancing in multi-task neural networks for mobile-network resource allocation and attention mechanisms for temporal prediction.

Industry Experience

Mcovision (independent), Machine Learning Engineer — Paris, France, 2021
- Built an explainable video classifier for rare heart disease and the full pipeline for data processing, model training, monitoring, and deployment.

SustainEcho, Research Intern — Paris, France, 2021
- Worked on meta-learning and few-shot personalization in low-data regimes, with model adaptation based on prototypical networks.

Education

PhD **University of Würzburg**, Reinforcement Learning — Germany, 2023 – 2027
- Expected April 2027. Supervisor: Prof. Carlo D'Eramo.
- Teaching: Introduction to Machine Learning; Introduction to Reinforcement Learning.

MSc **Université Paris-Saclay**, Optimization — France, 2021 – 2022
- Coursework: Convex analysis, compressed sensing, dynamic programming, tropical algebra, game theory, stochastic optimization and approximation.

MSc **ENS Paris-Saclay**, Mathematics, Vision, Learning (MVA) — France, 2020 – 2021
- Coursework: Convex optimization, reinforcement learning, kernel methods, computer vision, advanced deep learning, machine learning for text and graphs.

BSc **Télécom Paris / EURECOM**, Data Science — France, 2018 – 2020
- Coursework: Advanced statistical inference, statistical signal processing, machine learning, database management systems, cloud computing and security.

Supervision, Talks and Service

Supervision: Johannes Tölle, MSc thesis, University of Würzburg (2026). Recipient of the VDE Bayern Award 2026 and co-author of two papers above.

Talks: On Better Priors for Policy Search. Invited talk and lab visit, Improbable AI Lab (Pulkit Agrawal), MIT, August 2024.

Reviewer: NeurIPS, ICLR, ICML, RLC

Technical Skills

Computing: Python, JAX, PyTorch, NumPy, Julia, C

Methods: Policy optimization, Bayesian optimization, Gaussian processes, random matrix theory, spectral analysis

Tools: Ray, Weights & Biases, Git

Languages: Arabic (native), English (fluent), French (fluent), German (basic)